

Connective Tissue

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(cells, fiber, ground substanses)
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Connective Tissue

- Connective tissue is the most abundant and widely distributed tissue in the body .
- Connective tissue (CT) is the tissue that connect between all types of tissues in human body , and it has different functions
- Connective tissues can be rigid (bone), flexible (adipose), or fluid (blood).

Connective Tissue consists of three basic elements (**cells, ground substance, and fibers**)

1- Ground Substance

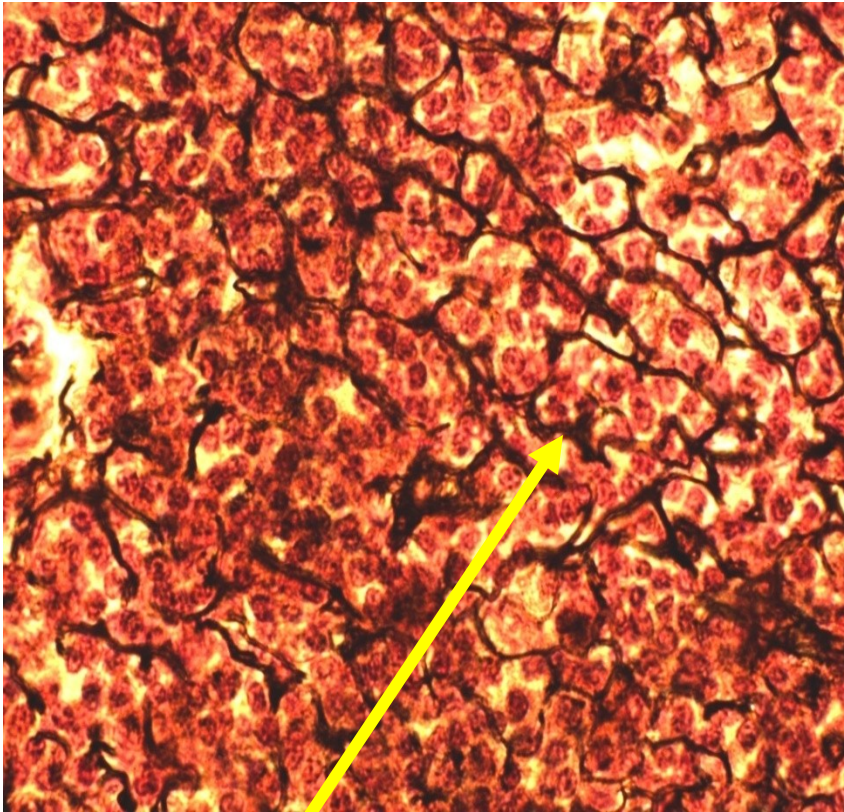
- The ground substance is gel like complex mixture, fills the spaces between cells and fibers.
- The ground substance holds large amounts of fluid and serves as a medium through which nutrients and other substances can diffuse between blood vessels and the cells.

Fibers -2

There are three types of fibers in Connective tissues

- A.) **collagen fibers** – are wide and wavy in appearance and generally stain pink. 79% of the protein in the body is collagen. provides tissues with tensile strength.
- B.) **elastic fibers** – are thin flexible fibers made from the protein elastin, that generally stain black. provide tissues with the ability to respond to stretch and distension
- C.) **reticular fibers** – Very thin and branched fibers. They have a spider web appearance and appear black under stain. provide a supporting framework for the cellular constituents of lymphoid organs mostly

Fiber Types



Reticular Fibers



Elastic Fiber

Collagen Fiber

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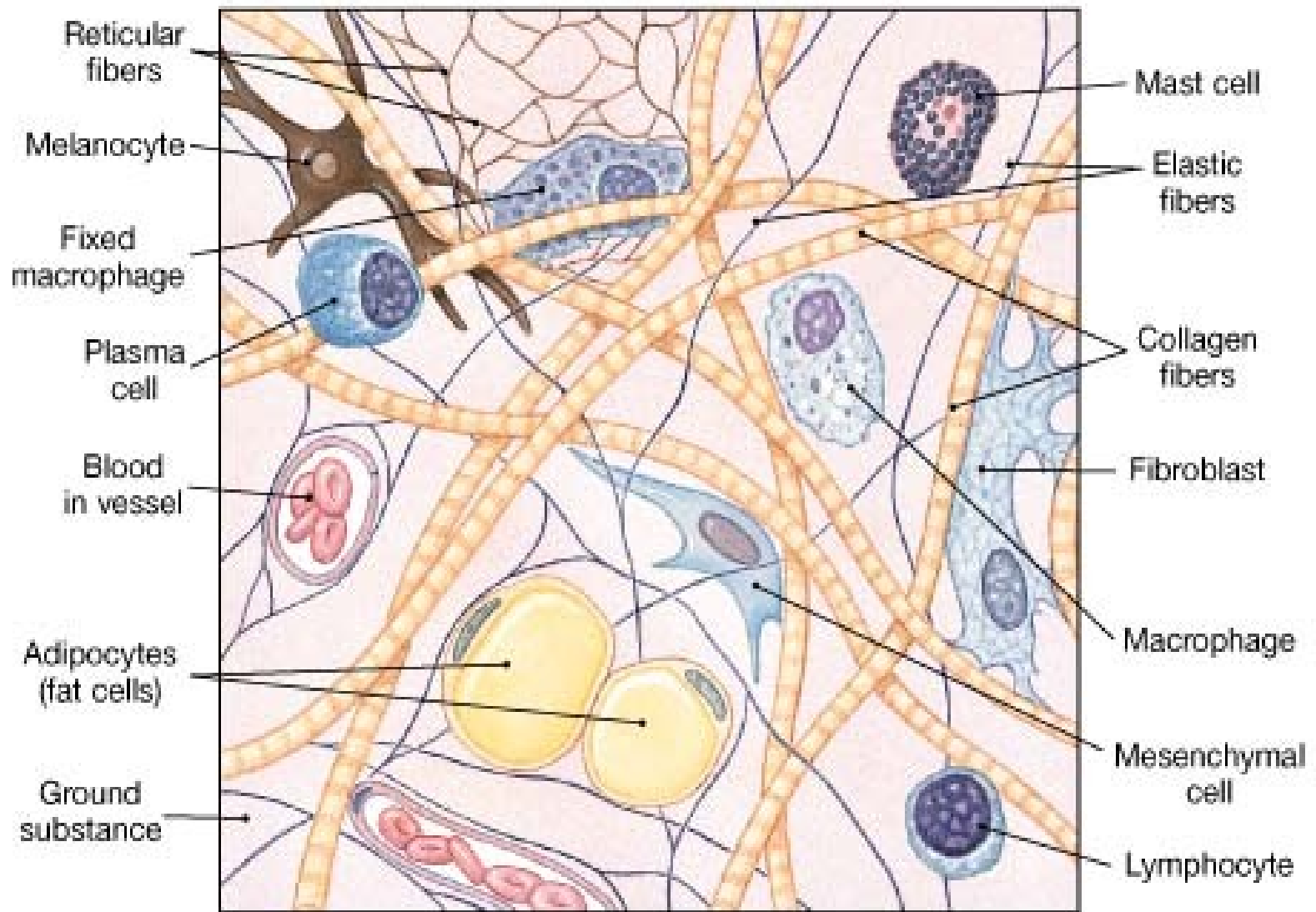
Connective tissue cells

fixed cell (permanent)

- Fibroblasts
 - Macrophages
 - Adipose cells
 - Mast cells
 - Reticular cells
 - Undifferentiated mesenchymal cells
-
- are native to tissues in which they are found.

Wandering cell (free)

- Lymphocytes
 - Plasma cells
 - Neutrophils
 - Eosinophils
 - Basophils
 - Monocytes
-
- are immigrant cells from blood or bone marrow



(a)

• **FIGURE 4-8 The Cells and Fibers of Connective Tissue Proper.** (a) A summary of the cell types and fibers of connective tissue proper. (LM $\times 384$)

Classification of Connective Tissue

1. Connective Tissue Proper.
- 2-Specialized Connective Tissues:
 - a. Fluid Connective Tissue
 - Blood
 - Lymph
 - b. Supporting Connective Tissue
 - Cartilage
 - Bone

Connective tissue proper includes:

1- Loose Connective tissue.

A- areolar connective tissue

B- adipose connective tissue

C- Reticular Connective tissue

2-Dense Connective tissue.

A- dense regular connective tissue

B- dense irregular connective tissue

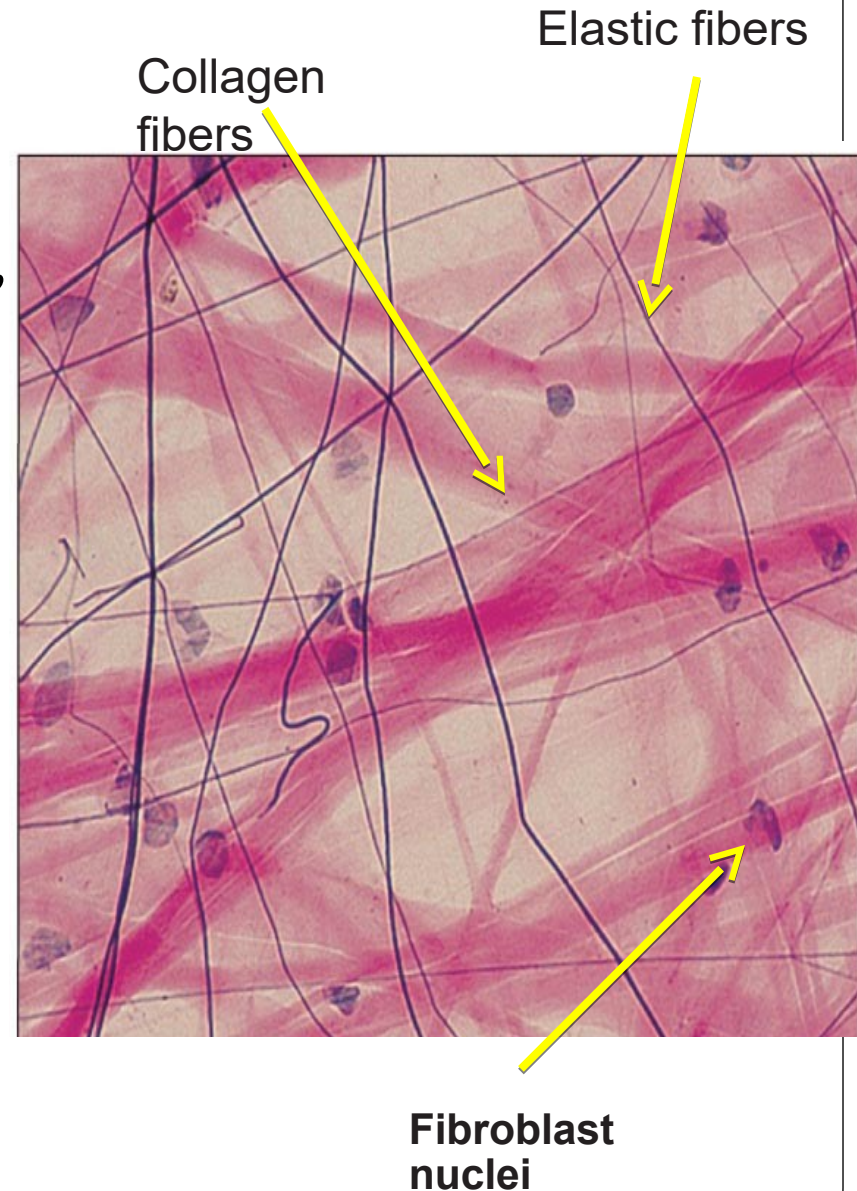
1- Loose Connective tissue.

A- areolar connective tissue

Gel-like matrix with all three fiber types; cells: fibroblasts, macrophages, mast cells, and some white blood cells.

Function: Wraps and cushions organs; its macrophages phagocytize bacteria; plays important role in inflammation; holds and conveys tissue fluid.

Location: Widely distributed under epithelia of body, e.g., forms lamina propria of mucous membranes; packages organs; surrounds capillaries.



.Figure 4.7 Areolar connective tissue: A prototype (model) connective tissue

Cell types

Macrophage

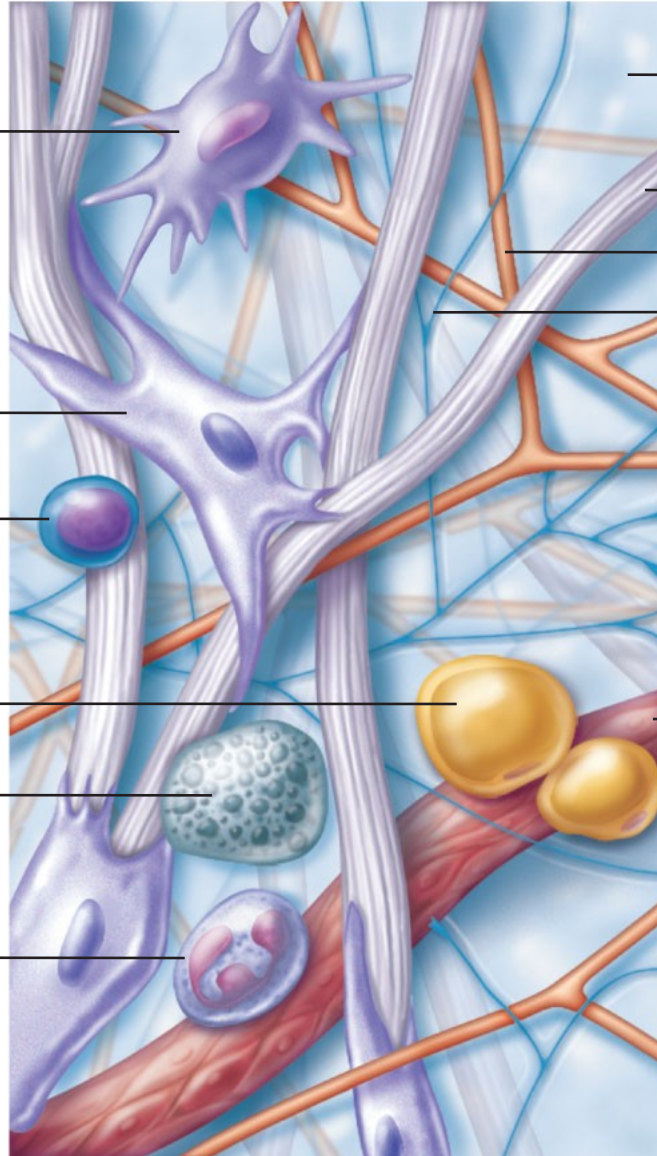
Fibroblast

Lymphocyte

Fat cell

Mast cell

Neutrophil



Extracellular matrix

Ground substance

Fibers

- Collagen fiber
- Elastic fiber
- Reticular fiber

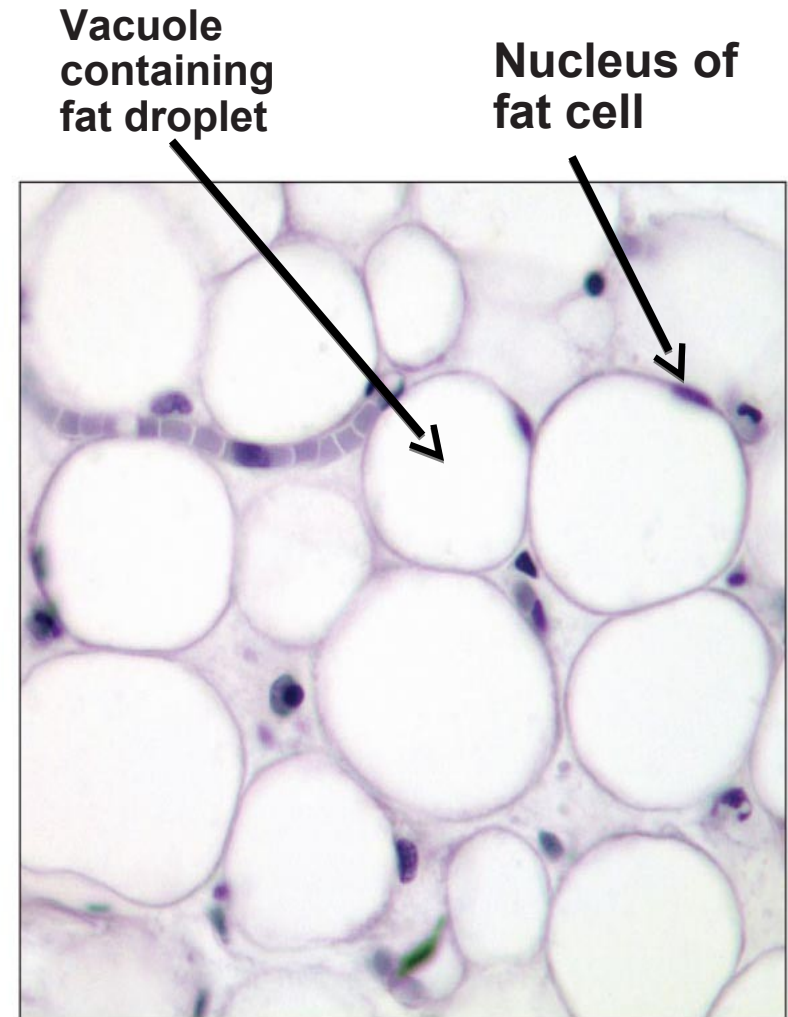
Capillary

B- adipose connective tissue

Matrix as in areolar, but very sparse; closely packed adipocytes, or fat cells, have nucleus pushed to the side by large fat droplet.

Function: Adipose tissue helps to cushion and protect organs, as well as insulate the body from heat loss.

Location: Under skin in the hypodermis; around kidneys and eyeballs; within abdomen; in breasts

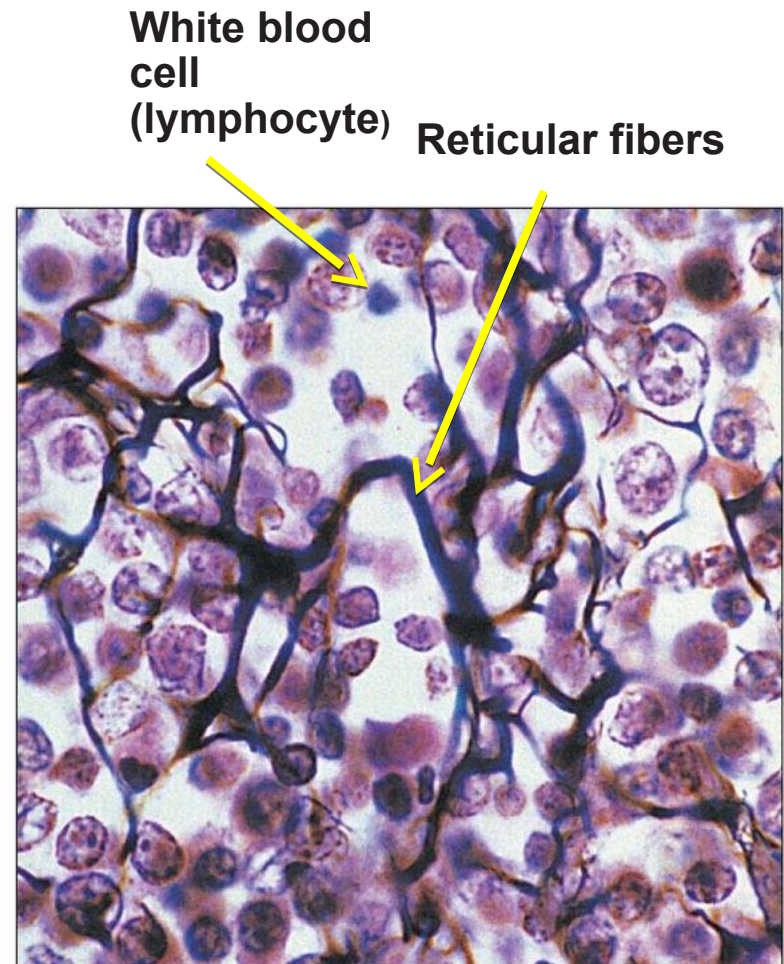


C- Reticular Connective tissue

Network of reticular fibers in a typical loose ground substance; reticular cells lie on the network.

Function: Fibers form a soft internal skeleton that supports other cell types including white blood cells, mast cells, and macrophages.

Location: Lymphoid organs (lymph nodes, bone marrow, and spleen).



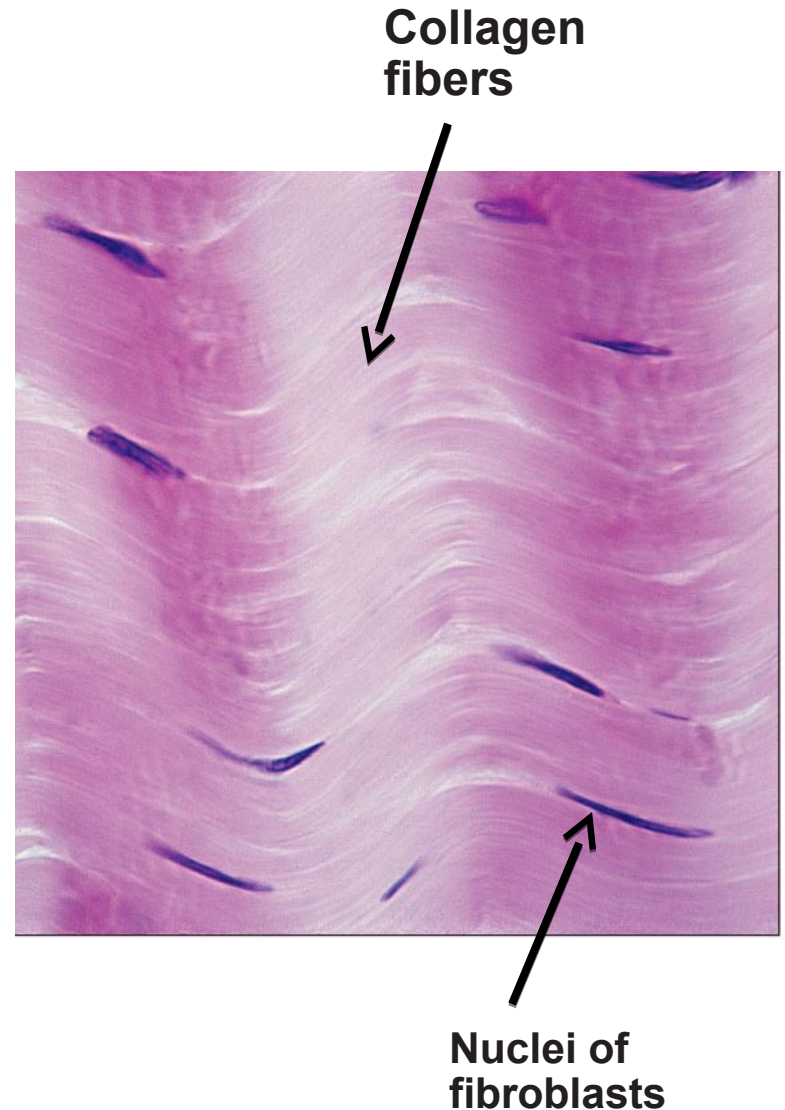
2-Dense Connective tissue.

A- dense regular connective tissue

Primarily parallel collagen fibers; a few elastic fibers; major cell type is the fibroblast.

Function: Attaches muscles to bones or to muscles; attaches bones to bones; withstands great tensile stress when pulling force is applied in one direction.

Location: Tendons, most ligaments.



B- dense irregular connective tissue

Primarily irregularly arranged collagen fibers; some elastic fibers; major cell type is the fibroblast.

Function: Able to withstand tension exerted in many directions; provides structural strength.

Location: Fibrous capsules of organs and of joints; dermis of the skin; submucosa of digestive tract.

